Unit - 1

1. Blockchain Basics

Blockchain kya hai?
Blockchain ek digital technology hai jisme data ko blocks ke form mein store kiya jata hai.
Har block previous block se connected hota hai, isliye ek chain banti hai.
Data ko network ke multiple computers par store kiya jata hai.
Ek baar data add hone ke baad usme change karna difficult hota hai.

Decentralized Ledger
Decentralized ledger ka matlab hai ki data kisi ek central server ke paas nahi hota.
Network ke multiple computers ke paas ledger ki copy hoti hai.
Har transaction ko network ke participants verify karte hain.
Isse single authority par dependency kam hoti hai.

Blockchain ke Uses
Blockchain ka use cryptocurrency jaise Bitcoin aur Ethereum mein hota hai.
Iska use secure transactions aur digital payments ke liye bhi kiya ja sakta hai.
Supply chain mein products ko track karne ke liye blockchain use hota hai.
Healthcare, voting aur digital identity jaise areas mein bhi iska use ho sakta hai.

2. Public Ledger

Public Ledger kya hai?
Public ledger ek aisa digital record hai jise network ke participants dekh aur verify kar sakte hain.
Blockchain mein transactions ka record ledger mein store hota hai.
Ye ledger kisi ek person ya organization ke control mein nahi hota.
Isse transactions ko track aur verify karna easy hota hai.

Transparency
Transparency ka matlab hai ki transactions ka record network participants ke liye visible hota hai.
Users transaction details ko verify kar sakte hain.
Isse hidden changes ya fraud ki possibility kam hoti hai.

Decentralization
Decentralization ka matlab hai ki ledger ka control kisi ek central authority ke paas nahi hota.
Multiple computers network par ledger ki copy maintain karte hain.
Isse ek single point of failure nahi hota.

Immutability
Immutability ka matlab hai ki blockchain mein record kiya gaya data easily change ya delete nahi kiya ja sakta.
Ek transaction block mein add hone ke baad usse modify karna difficult hota hai.
Isse records ki reliability aur trust improve hota hai.

Security

Public ledger mein transactions ko cryptography aur network verification se protect kiya jata hai.
Multiple computers transaction ko verify karte hain.
Isse unauthorized changes aur fraudulent transactions ko prevent karne mein help milti hai.

3. Bitcoin & Blockchain

Bitcoin blockchain ka use kaise karta hai
Bitcoin apne transactions ko record aur verify karne ke liye blockchain ka use karta hai.
Bitcoin network mein kisi central bank ki zarurat nahi hoti.
Har transaction blockchain par record hoti hai aur network participants usse verify karte hain.
Blockchain Bitcoin ke transactions ko secure aur transparent banata hai.

Transactions, Blocks aur Chain
Transaction ka matlab Bitcoin ko ek person se doosre person ko transfer karna hai.
Multiple verified transactions ko ek block mein collect kiya jata hai.
Ye blocks previous blocks se connected hote hain aur ek chain banate hain.
Isi connected structure ko **blockchain** kaha jata hai.

Bitcoin ke Benefits
Bitcoin ke through users directly ek-doosre ko digital money send kar sakte hain.
Isme traditional bank ya central authority ki direct requirement nahi hoti.
Transactions globally ki ja sakti hain aur blockchain par verify hoti hain.
Bitcoin limited supply ki wajah se digital asset ke roop mein bhi use hota hai.

4. Blockchain 2.0

Blockchain 1.0 vs 2.0
Blockchain 1.0 mainly **cryptocurrency aur digital payments** par focused tha.
Blockchain 2.0 ne blockchain ko programmable banaya, jisse applications aur smart contracts create kiye ja sake.
Bitcoin Blockchain 1.0 ka example hai, jabki Ethereum Blockchain 2.0 ka popular example hai.
Isliye Blockchain 2.0 ka scope cryptocurrency se kaafi broader hai.

Smart Contracts
Smart contract ek self-executing program hota hai jo blockchain par run karta hai.
Isme conditions pehle se define hoti hain aur condition fulfill hone par action automatically execute hota hai.
Example: payment receive hone par ownership automatically transfer ho jana.

dApps
dApps ka full form **Decentralized Applications** hai.
Ye applications blockchain network par operate karti hain aur usually kisi single central authority par depend nahi karti.
Ethereum jaise platforms par dApps develop ki ja sakti hain.

Ethereum
Ethereum ek blockchain platform hai jo cryptocurrency ke saath smart contracts support karta hai.

Iska native cryptocurrency **Ether (ETH)** hai.
Developers Ethereum par smart contracts aur dApps create kar sakte hain.
Ye Blockchain 2.0 ka important example hai.

Hyperledger
Hyperledger enterprise blockchain applications develop karne ke liye open-source blockchain projects ka ecosystem hai.
Ye mainly businesses aur organizations ke use cases ke liye designed hai.
Iska focus secure, permissioned aur business-oriented blockchain solutions par hota hai.

Blockchain 2.0 ke Applications
Blockchain 2.0 ka use **finance, supply chain, healthcare aur digital identity** jaise areas mein hota hai.
Smart contracts ke through agreements aur transactions automatically execute kiye ja sakte hain.
dApps ke through decentralized services create ki ja sakti hain.
Isse businesses mein transparency, automation aur trust improve kiya ja sakta hai.

5. Smart Contracts

Definition
Smart contract ek computer program hota hai jo blockchain par stored aur execute hota hai.
Isme agreement ki conditions pehle se code mein define hoti hain.
Conditions fulfill hone par contract automatically action perform karta hai.
Isme traditional middleman ki zarurat kam ho sakti hai.

Features
Smart contracts **automatic, transparent aur tamper-resistant** hote hain.
Ye blockchain network par execute hote hain aur predefined rules follow karte hain.
Contract ka execution network participants verify kar sakte hain.

Working
Sabse pehle contract ki conditions code mein define ki jati hain.
Contract blockchain par deploy kiya jata hai.
Jab required conditions fulfill hoti hain, contract automatically execute hota hai.
Execution ka result blockchain par record ho jata hai.

Real-life Example
Maan lo ek online seller aur buyer ke beech payment agreement hai.
Smart contract mein condition ho sakti hai: **"Payment receive hone ke baad product ownership transfer karo."**
Payment confirm hote hi contract automatically ownership transfer kar sakta hai.

Benefits
Smart contracts manual processes ko automate karte hain aur processing time reduce kar sakte hain.
Inme predefined rules hone ke karan human intervention kam hota hai.
Blockchain ki wajah se records transparent aur difficult to alter hote hain.
Isse transactions mein efficiency aur trust improve ho sakta hai.

6. Distributed Consensus

Consensus kya hai?
Consensus ka matlab hai network ke participants ka kisi transaction ya data ki validity par agree karna.
Blockchain mein central authority nahi hoti, isliye consensus important hota hai.
Ye ensure karta hai ki network mein correct aur valid data add ho.
Isse fake ya invalid transactions ko reject karne mein help milti hai.

Transaction Verification
Jab koi transaction hoti hai, network ke computers us transaction ko verify karte hain.
Wo check karte hain ki transaction valid hai aur sender ke paas required balance hai.
Valid transaction ko blockchain mein add karne ke liye process kiya jata hai.
Invalid transaction ko network reject kar deta hai.

Proof of Work (PoW)
Proof of Work ek consensus mechanism hai jisme miners complex mathematical problem solve karte hain.
Jo miner problem successfully solve karta hai, woh new block add karne ka opportunity paata hai.
Is process mein significant computing power aur electricity use hoti hai.
Bitcoin PoW consensus mechanism use karta hai.

Proof of Stake (PoS)
Proof of Stake mein participants apne cryptocurrency tokens ko stake karke transaction validation mein participate karte hain.
Network selected validators ke through new blocks validate karta hai.
PoW ke comparison mein PoS ko generally less computational energy ki zarurat hoti hai.
Ethereum ab Proof of Stake use karta hai.

Bitcoin mein Consensus
Bitcoin transactions ko validate karne ke liye **Proof of Work (PoW)** use karta hai.
Miners transactions ko collect karke blocks banate hain aur mathematical puzzle solve karte hain.
Valid block blockchain mein add hota hai aur network usse accept karta hai.
Is process se Bitcoin network bina central authority ke consensus achieve karta hai.

7. Blockchain Chain & Longest Chain

Blockchain chain kya hai?
Blockchain chain interconnected blocks ka sequence hota hai.
Har new block previous block se cryptographically linked hota hai.
Is chain mein transactions ka chronological record maintain hota hai.
Isliye blockchain ko ek continuously growing chain of blocks kaha jata hai.

Multiple chains kaise banti hain
Kabhi-kabhi network ke different participants same time par different blocks create kar dete hain.
Isse blockchain temporarily do ya more branches mein divide ho sakti hai, jise **fork** kaha jata hai.
Network consensus ke through eventually ek branch ko continue karta hai.
Dusri branch usually discarded ya orphaned ho jati hai.

Longest Chain Rule

Longest Chain Rule ke according network generally us chain ko valid chain maanta hai jisme sabse zyada cumulative Proof of Work hota hai.
Agar multiple valid chains temporarily exist karti hain, nodes stronger chain ko follow karte hain.
Isse network mein ek consistent blockchain maintain karne mein help milti hai.
Bitcoin mein ye rule consensus process ka important part hai.

8. Permissioned Blockchain

Permissioned vs Public Blockchain
Public blockchain mein koi bhi person network join karke transactions participate kar sakta hai.
Permissioned blockchain mein network access sirf authorized users ko diya jata hai.
Public blockchain generally open aur decentralized hoti hai, jabki permissioned blockchain controlled access provide karti hai.
Example: Bitcoin public blockchain hai, jabki Hyperledger Fabric permissioned blockchain ka example hai.

Permissioned Model
Permissioned model mein users ko blockchain access karne ke liye permission ya authorization ki zarurat hoti hai.
Organization decide karti hai ki kaun data read, write ya validate kar sakta hai.
Isme identities ko manage aur participants ko control kiya ja sakta hai.
Ye model businesses ke liye useful hota hai jahan privacy important hoti hai.

Business Use
Permissioned blockchain ka use banks, supply chains, healthcare aur insurance jaise industries mein ho sakta hai.
Companies iska use transactions aur records ko securely share karne ke liye kar sakti hain.
Authorized organizations ek common ledger maintain kar sakti hain.
Isse transparency, data sharing aur business process efficiency improve ho sakti hai.

9. Cryptographic Hash Function

Hash Function kya hai?
Hash function ek mathematical function hai jo input data ko ek fixed-length output mein convert karta hai.
Is output ko **hash** ya **hash value** kaha jata hai.
Same input dene par same hash milta hai.
Blockchain mein hashes data ko secure aur blocks ko link karne ke liye use hote hain.

Deterministic
Deterministic ka matlab hai ki same input ke liye hamesha same hash output milega.
Example: `"Hello"` ko hash karne par har baar same hash generate hoga.
Input mein ek small change hone par output completely different ho sakta hai.

Fixed Size
Hash function input chahe chhota ho ya bahut bada, output fixed size ka hota hai.
Example: SHA-256 hamesha **256-bit hash** produce karta hai.
Isse blockchain mein hash values ko efficiently store aur compare karna possible hota hai.

Fast Computation
Hash function ko input data se hash generate karne mein relatively kam time lagta hai.
Computers large amount of data ka hash quickly calculate kar sakte hain.
Blockchain mein transaction aur block verification ke liye fast hashing important hai.

Preimage Resistance
Preimage resistance ka matlab hai ki hash value se original input ko find karna computationally very difficult hota hai.
Agar kisi ko sirf hash pata hai, to original data easily reverse nahi kiya ja sakta.
Ye property blockchain data ki security improve karti hai.

Collision Resistance
Collision tab hota hai jab do different inputs ka same hash generate ho.
Collision resistance ka matlab hai ki aise two inputs find karna extremely difficult hona.
Isse hash-based security aur blockchain records ki reliability maintain hoti hai.

Avalanche Effect
Avalanche effect ka matlab hai input mein bahut small change karne par hash mein bahut large change ho jana.
Example: `"Hello"` aur `"hello"` ke hashes completely different honge.
Isse data mein even small modification easily detectable ho sakta hai.

10. Hash Pointer & Merkle Tree

Hash Pointer
Hash pointer ek reference hota hai jo kisi data/location ke saath us data ka cryptographic hash store karta hai.
Isse data ki location ke saath uski integrity bhi verify ki ja sakti hai.
Agar original data change ho jaye, to hash bhi change ho jayega.
Isliye hash pointer tampering detect karne mein useful hai.

Blockchain mein Hash Pointer
Blockchain mein har block previous block ke hash ko reference karta hai.
Isse blocks ek doosre se securely connected rehte hain.
Agar kisi old block ka data change kiya jaye, to uska hash change ho jayega.
Is change ki wajah se next blocks ke references mismatch ho jayenge.

Merkle Tree
Merkle Tree ek tree-like structure hai jisme multiple transaction hashes ko efficiently organize kiya jata hai.
Neeche transaction hashes hote hain aur unhe pair karke naye hashes generate kiye jate hain.
Ye process continue hota hai jab tak ek single hash remaining nahi rehta.
Ye single hash **Merkle Root** hota hai.

Merkle Root
Merkle Root Merkle Tree ka top-level single hash hota hai.
Ye block ke multiple transactions ko ek single hash se represent karta hai.
Agar kisi transaction mein change ho, to uska hash aur eventually Merkle Root change ho jayega.
Isliye Merkle Root transaction integrity verify karne mein useful hai.

Transaction Verification
Merkle Tree se kisi specific transaction ko poori list check kiye bina verify kiya ja sakta hai.
Transaction ke hash ke saath required sibling hashes ko use karke Merkle Root calculate kiya jata hai.
Agar calculated root block ke stored Merkle Root se match karta hai, transaction valid maana ja sakta hai.

4 Transactions ka Merkle Tree Example
Maan lo 4 transactions hain: **T1, T2, T3, T4**.
Pehle unke hashes banenge: **H1, H2, H3, H4**.
Phir **H1+H2 H12** aur **H3+H4 H34** generate honge.
Finally **H12+H34 Merkle Root** generate hoga.

```
         Merkle Root
         /           \
      H12           H34
     /    \          /    \
   H1    H2       H3    H4
    |      |          |       |
   T1    T2      T3    T4
```